Huu-Truong Le

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Education

- **Université de Lorraine** — Nancy, France
  PhD in Automatic Control, Signal and Image Processing, Computer Engineering — *07/2023 – present*
  - **Thesis title**: AI-based machine learning and reasoning for predictive maintenance in Industry 4.0
- **University of Danang - University of Science and Technology** — Danang, Vietnam
  Engineer's degree in Automatic Control — *09/2018 – 09/2022*

Research Experience

- **Nancy Research Center for Automatic Control** — Nancy, France
  PhD student — *07/2023 – present*
  - PhD research on adaptive maintenance scheduling for complex multi-component systems, with a focus on reconfigurable manufacturing systems, structural dependencies, and disassembly-induced degradation.
  - Developed maintenance optimization frameworks combining dynamic grouping, continuous/discrete degradation modeling, and rolling-horizon decision-making to support cost-effective and reliable maintenance planning.
  - Applied graph-based deep reinforcement learning using Graph Neural Networks to optimize structure-aware maintenance decisions.
- **University of Danang - University of Science and Technology** — Danang, Vietnam
  Teaching assistant — *08/2021 – 12/2021*
  - Mentored students through project-based learning, supporting hardware and software design and implementation.

Working Experience

- **Ban Vien Corporation** — Danang, Vietnam
  Embedded Software Engineer — *10/2022 – 02/2023*
  - Developed a SystemC/C++ model of an automotive System-on-Chip (SoC) to support system-level simulation, performance evaluation, and verification of hardware/software interactions during early design stages.
- **Thua Thien Hue Power Company** — Hue, Vietnam
  Intern — *04/2022 – 06/2022*
  - Developed practical knowledge of SCADA systems and the IEC 61850 standard through the survey and analysis of the Phu Bai substation, contributing to the evaluation of substation monitoring, control, and communication solutions.

Projects

- **MODAPTO** [**website**] — 07/2023 – 01/2026
  Repository
  - Contributed to Work Package 5 of the project, focusing on predictive maintenance services and maintenance decision-support for industrial systems.
  - Analyzed industrial data provided by the project partner (SEW USOCOME) to support degradation assessment, maintenance planning, and decision-making requirements.
  - Developed a dynamic grouping maintenance policy to optimize predictive maintenance interventions and minimize total maintenance costs while supporting reliable system operation.

Publications

- **Journals**
  - **Huu-Truong Le**, Phuc Do & Alexandre Voisin. "Deep graph reinforcement learning–based maintenance scheduling for multi-component systems considering disassembly impact" . *Reliability Engineering and Systems Safety.* [To be submitted in August 2026]
  - **Huu-Truong Le**, Phuc Do, Alexandre Voisin, & Chiara Franciosi. "Maintenance Scheduling in Reconfigurable Manufacturing Systems: A Literature Review and a Dynamic Grouping–Based Framework . *Journal of Risk and Reliability.* [under revision R1]

- **Conferences**
  - **Huu-Truong Le**, Phuc Do, Alexandre Voisin, & Chiara Franciosi. "AI-based maintenance scheduling framework considering disassembly impact". 23rd IFAC World Congress, 2026. [accepted]
  - **Huu-Truong Le**, Phuc Do, Alexandre Voisin, & Chiara Franciosi. "A Framework for Maintenance Scheduling in Reconfigurable Manufacturing Systems". 13th IMA International Conference on Modelling in Industrial Maintenance and Reliability, 2025.
  - Huy Vu Tran, Kim Anh Nguyen, Duc Hanh Dinh, **Huu-Truong Le**, & Phuc Do. "Evaluation of the health state of oil-immersed transformer based on dissolved gas analysis: An empirical study". 13th IMA International Conference on Modelling in Industrial Maintenance and Reliability, 2025.
  - **Huu-Truong Le** , Chiara Franciosi, Phuc Do, & Alexandre Voisin. "Contribution of Maintenance to Reconfigurable Manufacturing Systems: State of the Art and Challenges". 6th IFAC Workshop on Advanced Maintenance Engineering, Services and Technology AMEST, 2024.

Courses and certifications

- **Courses**: Machine Learning, Reinforcement Learning, Graph Neural Network [1] [2]

Skills

- **Languages**: English (C1), French (Beginner), Vietnamese (Native)
- **Programming languages**: Python, C/C++, Matlab, SQL

References

- **Phuc Do** Email : phuc.do@mines-ales.fr
  Professor - IMT Mines Alès
- **Alexandre Voisin** Email : alexandre.voisin@univ-lorraine.fr
  Professor - Université de Lorraine